

Viktória Kecskeméthy

Research Interests

- Atmospheres of brown dwarfs and (exo)planets
- Machine learning applications in astronomy

Education

2024 – **PhD in Astronomy**, University of Hertfordshire, UK

Project: *Remote Sensing Alien Clouds and Chemistry with JWST*

Supervisors: Dr. Ben Burningham, Dr. Fei Wang, Prof. Hugh Jones

- Performing retrievals and atmospheric characterization of brown dwarfs to analyze their clouds, chemistry, and thermal structure, using JWST data.
- Implementing Neural Posterior Estimation (NPE) into the [Brewster](#) atmospheric retrieval code to enhance the speed of parameter distribution retrieval.

2021 – 2023 **MSc in Astronomy Research**, Leiden University, Netherlands, Grade: 7.5/10 (US: A-, UK: Upper Second Class)

2nd MSc Thesis: *Impact of Thermal Evolution on Exoplanetary Atmospheres*

Supervisors: Dr. Yamila Miguel, Amy Louca

- Modeled the thermal evolution and compositional changes of various H₂-dominated exoplanets.
- This involved utilising the disequilibrium chemistry code VULCAN and radiative transfer code petitRADTRANS.
- Further proved that the CO₂ feature at $\sim 4.3 \mu\text{m}$ in transmission spectra is highly sensitive to thermal evolution, providing insights into planet formation and evolution with instruments like JWST.

1st MSc Thesis: *Substructure of Planet-Forming Disks with VLT*

Supervisors: Prof. Dr. Michiel Hogerheijde, Dr. József Varga

- Conducted statistical analysis on the inner disk structures of planet-forming disks observed with VLT instruments.
- Fitted geometric models to determine disk size and structure across different wavelengths.
- Detected inner-disk gaps indicating potential nascent planets around CQ Tau, HD 100546, HD 142527, and HD 144432.

2020 – 2021 **MSc in Astronomy**, Eötvös Loránd University, Hungary, *Discontinued, went to Leiden instead*

2017 – 2020 **BSc in Earth Sciences**, Eötvös Loránd University, Hungary, Grade: 5.0/5.0 (US: A+, UK: First Class)

BSc Thesis: *Lightcurves of Hilda Asteroids and Transneptunian Objects*

Supervisor: Dr. Csaba Kiss

- Developed shape models of asteroids from scratch using Python, based on their light curves.
- Fitted these models to Kepler K2 data to determine shapes, rotation periods, and densities.
- Identified 15 contact binary candidates among 155 Hilda asteroids.

Appointments

Nov 2024 – **Visiting Lecturer**, Department of Physics, Astronomy and Maths, University of Hertfordshire, UK

Present ○ Leading PC laboratories for Data Science MSc students.

Sep 2022 – **Teaching Assistant**, Leiden University, Faculty of Science, Netherlands

Dec 2022 ○ Led practical sessions and graded weekly assignments for the mandatory Astronomy BSc course, 'Stars'.

Oct 2019 – **Observatory Assistant**, Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences,

Aug 2021 Hungary

- Performed observations at Piszkestető Mountain Station (3-4 nights every month).
- Assisted in data reduction and analysis of the observed data.

Publications (as a first author)

1. **Kecskeméthy, V.**; Louca, A.; Miguel, Y. (2024). *Temporal Variability in Transmission Spectra of H₂-Dominated Exoplanets: The Influence of Thermal Evolution and Stellar Irradiation on Atmospheric Composition*. *New Astronomy*, Volume 113, id.102296, December 2024. [\[ADS\]](#)
2. **Kecskeméthy, V.**; Kiss, C., et al. (2023). *Light Curves of Transneptunian Objects from the K2 Mission of the Kepler Space Telescope*. *The Astrophysical Journal Supplement Series*, Volume 264, Issue 1, id.18, 20 pp., January 2023. [\[ADS\]](#)

Publications (as a co-author)

1. Varga, J.; [and 39 others, including **Kecskeméthy, V.**] (2024). *Mid-infrared Evidence for Iron-Rich Dust in the Multi-Ringed Inner Disk of HD 144432*. *Astronomy & Astrophysics*, Volume 681, id.A47, 29 pp., January 2024. [\[ADS\]](#)
2. Csörnyei, G.; [and 9 others, including **Kecskeméthy, V.**] (2022). *Study of Changes in the Pulsation Period of 148 Galactic Cepheid Variables*. *Monthly Notices of the Royal Astronomical Society*, Volume 511, Issue 2, pp.2125-2146, April 2022. [\[ADS\]](#)
3. Marciniak, A.; [and 67 others, including **Kecskeméthy, V.**] (2021). *Properties of Slowly Rotating Asteroids from Convex Inversion Thermophysical Model*. *Astronomy & Astrophysics*, Volume 654, id.A87, 32 pp., October 2021. [\[ADS\]](#)
4. Kalup, C.; [and 10 others, including **Kecskeméthy, V.**] (2021). *101 Trojans: A Tale of Period Bimodality, Binaries, and Extremely Slow Rotators from K2 Photometry*. *The Astrophysical Journal Supplement Series*, Volume 254, Issue 1, id.7, 20 pp., May 2021. [\[ADS\]](#)

Conferences (presented or contributed)

1. The inner region of planet forming disks from the VLT/MATISSE GTO YSO survey ([ADS](#))
Varga, József; [and 38 others, including **Kecskeméthy, Viktória**]
European Astronomical Society Annual Meeting 2024
2. Rotational properties of Kuiper belt objects as seen by the K2 mission ([ADS](#))
Kecskeméthy, Viktória; Kiss, Csaba, et al.
16th Europlanet Science Congress 2022
3. Rotational properties of transneptunian objects from the K2 mission ([ADS](#))
Kiss, Csaba; **Kecskeméthy, Viktória**, et al.
14th Europlanet Science Congress 2020
4. New census of Hilda, Jovian Trojan, Centaur, and transneptunian object light curves from K2 measurements ([ADS](#))
Kiss, Csaba; [and 15 others, including **Kecskeméthy, Viktória**]
EPSC-DPS Joint Meeting 2019

Talks & Posters

1. **Investigating Silicate Clouds in Benchmark L-T Dwarfs with JWST** ([Poster](#))
European Astronomical Society Annual Meeting 2025 Cork, Ireland
2. **SDSSJ141659.78+500626.4: Investigating Silicate Clouds on an L dwarf** (Talk)
UH-PAM Lunch talk 2025 Hatfield, UK
3. **NH₃ observations of the Orion Molecular Cloud** (Talk)
Interaction of Stars with their Environment 2019 Budapest, Hungary

Conferences & workshops (attended)

1. DISCnet Data Analysis and Statistics Workshop, March 2025 London, UK
2. GRADnet Machine Learning & AI Workshop, Nov 2024 London, UK
3. GRADnet Induction Day, Oct 2024 (themes: *Machine Learning* and *Publication & Peer Review*) London, UK

Scholarships & Awards

- 2024 – 2028 **Science and Technology Facilities Council (STFC) PhD Studentship**, University of Hertfordshire
- Spring 2020 **Research Scholarship (tudományos ösztöndíj)**, Eötvös Loránd University
- Dec 2019 **Young Scientists' Conference (TDK)**: Submitted 2 projects, awarded 1st and 2nd place, Eötvös Loránd University
- Spring 2018 – Fall 2019 **Academic Excellence Scholarship (tanulmányi ösztöndíj)** won it four times, Eötvös Loránd University

Technical skills

- Advanced **Python** (NumPy, Matplotlib, Pandas, SciPy, Astropy, PyTorch etc.), **LaTeX** (using it for reports, publications, and CVs)
- Beginner IDL, Fortran, SQL, R, C, C#